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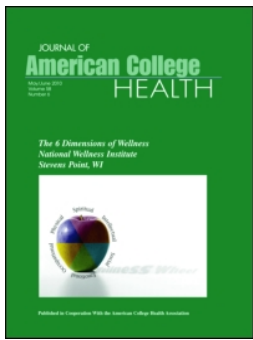


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MAJOR ARTICLE



When culture and health collide: feminine honor endorsement and attitudes toward catch-up HPV vaccinations in college women

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ABSTRACT

Objective: Sexual promiscuity stigma constitutes a significant barrier which prevents young women from receiving catch-up human papillomavirus (HPV) vaccinations. Our objective was to explore how this stigma may be particularly detrimental to women in cultures of honor, where women are expected to maintain a reputation of sexual purity.

Participants: Participants were a sample of 182 young women who had not previously received an HPV vaccination.

Methods: We measured feminine honor endorsement, HPV sexual promiscuity stigma, HPV vaccination shame, HPV vaccination intentions, and a set of covariates.

Results: Analyses revealed that the significant association between feminine honor endorsement and HPV vaccination intentions is fully explained by HPV sexual purity stigma and HPV vaccination shame.

Conclusions: These findings indicate that honor-endorsing women are significantly impacted by sexual purity stigma, minimizing intentions to receive HPV vaccinations. These findings also highlight the need for the inclusion of cultural factors in the study of HPV vaccine uptake.

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According to the Center for Disease Control (CDC), approximately 79 million Americans are currently infected with HPV (HPV), with the majority of these cases occurring in individuals in their late teens and early 20's.¹ Considering that there are currently no approved HPV screening tests for men, previous research efforts have been primarily focused on prevention and detection of HPV infections and HPV-related diseases in women. This research has shown that HPV infection rates are particularly high in young women, with peak infection rates falling between 18 and 25 years old², which validates the prior research's strong emphasis on studying college-aged women due to being in an age group disproportionately affected by HPV.^{3,4} The risks of such infections are apparent given the cancerous outcomes associated with HPV infection. Approximately 93% of all women's cancers in sites where HPV can be found are attributed to an HPV infection of some strain.⁵ With these risks in mind, the CDC recommends girls to receive the HPV vaccination early in life, preferably between the ages of 9 and 12, so as to protect them against HPV infection before they become sexually active.⁶ Unfortunately, HPV vaccination rates among these young girls remain relatively low in the United States. For instance, the latest National Immunization Survey (NIS) reported that in 2018 the percentages of adolescents aged 13–17 years who were fully vaccinated for HPV ranged from 39.9% to 57.1%, with an average of 51.1% (Walker et al., 2019⁷). Although these rates

are slightly improved compared to 2017's estimates (48.6%), this improvement is mainly attributed to the increase of vaccinations among males (i.e., from 44.3% to 48.7%) while vaccination rates among females remained relatively constant (from 53.1% to 53.7%). Among these young girls, both the low HPV vaccination rates and the lack of increase in HPV vaccination rates over time are concerning. Importantly, it highlights the importance of emphasizing a catch-up of HPV vaccination in young adult women.

A catch-up HPV vaccination is recommended for individuals between the ages of 13 and 26 years who have not been vaccinated previously or who have not completed the vaccination series that consist of three doses administered at 0, 2, and 6 months.⁸ This catch-up HPV vaccination has been recommended since 2006 by the CDC Advisory Committee on Immunization Practices.⁸ Although the HPV vaccine is most effective when given at a younger age (9 to 12 years old), presumably before exposure to any HPV through sexual contact occurs, the vaccine may have substantial health benefits when administered to young adult women who are 26 years or younger.⁸ Importantly, research has shown that HPV vaccination is not only cost-effective for young girls in the CDC-recommended 9–12-year-old age range, but also for young adult women up to the age of 25 years.⁹ Similarly, the ACIP suggests that catch-up HPV vaccinations can be offered to adult women aged 27 through 45 years who are not adequately vaccinated or at low risk

of prior HPV exposure (e.g., limited number of previous sexual partners, no previous sexual activity), but are likely to have a future risk of HPV exposure.^{8,10} Importantly, a number of clinical trials have shown that HPV vaccination for women older than 25 years is effective and safe.^{11–14} Accordingly, catch-up HPV vaccinations should continue to be a top priority for young adult women and, if necessary, for older adult women who are still at risk for future HPV exposure.

Despite the importance of catch-up HPV vaccinations, it appears that very few young adult women heed the recommendation of getting them. For example, a study by Williams and colleagues¹⁵ found that only 23.2% of unvaccinated women aged 18–26 had initiated the HPV vaccine sequence. A more recent study found that the vaccination rate for HPV among 18–26-year-old women had increased since 2013, but only to 35.3%.¹⁶ These low rates suggest that it is imperative to study not only the reasons why young girls are not receiving the HPV vaccine, but also to study which factors may prevent young adult women from receiving the vaccination as well.

What factors might be preventing traditional college-age women (26 years old or younger) from initiating the vaccine sequence? Prior research has helped to identify barriers keeping women from receiving an HPV vaccination, such as perceptions regarding the safety and effectiveness of the vaccination^{15,17–19}, perceived risk of contracting HPV^{20–23}, frequency of sexual activity^{24–27}, and prior knowledge about HPV.^{28–29} Of interest to the current study, another significant barrier to young women seeking out HPV vaccinations is stigma.

Stigma surrounding HPV vaccinations for young women appears to be centered around beliefs that those who receive the HPV vaccination are sexually promiscuous or “unclean”. For example, Hopfer and Clippard³⁰ found that, in a sample of 20-year-old women, a recurring theme shared by participants was that only women who were sexually promiscuous would need to receive the HPV vaccine, with one participant even indicating that “only sluts get HPV” (p. 271). Similarly, Wang and colleagues³¹ found that 64% of participants considered “promiscuous people” as the group most needing an HPV vaccination (p. 3241). Whereas, Pierre-Victor and colleagues³² found that some women believed that HPV vaccines were mainly intended as a “preventive measure for the sexually promiscuous” (p. 93). These sentiments were mirrored by young women in other studies as well.^{33–35} Unfortunately, the stigma attached to HPV appears to create a strong barrier which prevents young women from seeking out HPV vaccinations. Consistent with this view, Krakow and colleagues³⁶ found that beliefs that receiving an HPV vaccination would create a reputation of being sexually promiscuous was associated with decreased intention of receiving an HPV vaccination.

Considering the aforementioned findings, one might suspect that women who express deep concern for their social reputation should experience the strongest apprehension regarding the stigma attached to HPV vaccination. One such group of women may be found in cultures of honor. Cultures

of honor are characterized by an ideology which places high importance on reputation defense and reputation management, and are found in many areas of the world including the American South, in the Mediterranean, and in the Middle East.^{37–43} For men in cultures of honor, their reputation is centered around being seen as strong and fearless, even if doing so requires forms of retributive violence. In fact, masculine honor has been used to explain homicide rates, reactions to insult, and even school violence.^{44–47} For women, however, one’s social reputation is primarily concerned with sexual purity, chastity, and fidelity.^{48–49} Thus, honor-endorsing women should do whatever is necessary to avoid being seen as sexually promiscuous or sexually “impure”.^{48,50–53} This feminine facet of honor is seen as so important to honor endorsers that any violation of these norms, such as acts of sexual infidelity, are oftentimes met with some form of aggression, such as domestic violence or sexual aggression.^{54–56} Accordingly, one should expect that women who endorse feminine honor ideology would be particularly vigilant to instances that may render them a social reputation of sexual promiscuity. Therefore, we would expect that honor-endorsing women in the American South, with a heightened vigilance toward threats to social reputation^{46,57}, should face a particularly strong barrier in the “catch-up” vaccination phase due to the association between sexual promiscuity stigma and HPV vaccinations. This possibility poses a widespread threat to women’s health in the American South considering that the traditional “honor belt” in the United States stretches across 27 states.⁵⁸

Based on prior research focusing on women’s negative perceptions of HPV vaccinations and research suggesting the importance of sexual purity in honor-endorsing women, the goal of the current study is to examine the impact of feminine honor endorsement on intentions to seek catch-up HPV vaccinations among young adult women. Our hypotheses will test two assumptions. First, we will test the hypothesis that feminine honor endorsement is associated with decreased likelihood seek out an HPV vaccination.

Hypothesis 1: Feminine honor endorsement will be associated with lower HPV vaccination intentions in a sample of unvaccinated young women.

Second, we will test the hypothesis that the link between feminine honor endorsement and HPV vaccination intentions is explained by sexual purity stigma and the anticipated shame and embarrassment a woman would experience if she were to receive a vaccination. Sexual purity stigma will refer to the negative attitudes women may hold about how receiving a vaccination implies sexual promiscuity—this form of stigma has been referred to elsewhere as public stigma.³⁰ Anticipated shame and embarrassment is distinct from the sexual purity stigma variable, as it will measure the anticipated affective response to receiving a vaccination, not necessarily cognition regarding the vaccine. This distinction between stigma and shame has been utilized in prior health research as well.⁵⁹

Hypothesis 2: The link between feminine honor endorsement and HPV vaccination intentions will be mediated by

sexual purity stigma and anticipated shame and embarrassment one would feel upon receiving the vaccination.

Methods

Participants

357 undergraduate women were sampled from a Southwestern university, volunteering to take a study on “health and social attitudes” for partial completion of their research exposure requirement in their Introductory Psychology course. Of the full sample, 182 women (50.9%; $M_{\text{age}} = 18.82$) indicated that they had not previously received the HPV vaccine, which was the portion of the sample that was used for subsequent analyses. The final sample was predominately white (68.7%), with the remainder of the sample identifying as Latino/Hispanic (8.2%), Asian (8.2%), African American (5.5%), Native American (4.4%) or “Other” (5.0%). This study received approval from the institutions Institutional Review Board (IRB), and all participants agreed to an informed consent document prior to beginning the study.

Measures

Honor ideology for womanhood scale (HIW; $\alpha = .70$)

As a measure of feminine honor endorsement, the 12-item HIW was administered.⁶⁰ The HIW addresses core aspects of feminine honor endorsement, with items such as “A good woman never flirts with a man who is not her husband or boyfriend,” and “A respectable woman never wants to be known as being sexually permissive”. Participants responded on a scale from 1 (*Strongly Disagree*) to 7 (*Strongly Agree*), and the items were averaged to create a summarized score for each participant.

HPV sexual purity stigma, (HPV-Stigma; $\alpha = .92$)

A 5-item HPV-Stigma scale was created to assess the beliefs that receiving the HPV vaccination would imply a person is “unclean” or sexually promiscuous. Two representative items in this scale are: “Receiving the HPV vaccine probably means you are sexually promiscuous” and “People who get the HPV vaccine probably have many sexual partners”. Participants responded on a scale from 1 (*Strongly Disagree*) to 7 (*Strongly Agree*), and the items were averaged to create a summarized score for each participant.

HPV embarrassment and shame. (HPV-Shame; $\alpha = .94$)

A 4-item HPV-Shame scale was created to assess the extent to which women would be ashamed or embarrassed if people knew they had received the HPV vaccine. Items included “I would be embarrassed if people knew I had gotten the HPV vaccine” and “It would make me feel ashamed if people knew I had received the HPV vaccine”. Participants responded on a scale from 1 (*Strongly Disagree*) to 7 (*Strongly Agree*), and the items were averaged to create a summarized score for each participant.

HPV vaccination intentions (HPV-Intent); $\alpha = .97$

A 3-item HPV-Intent scale was created to assess women’s intentions to receive the HPV vaccine. Items included “To what extent would you be willing to get the HPV vaccination?” and “To what extent would you be willing to get the HPV vaccination in an upcoming doctors visit?”. Participants responded on a scale from 1 (*Definitely Would Not*) to 7 (*Definitely Would*), and the items were averaged to create a summarized score for each participant.

Covariates

Several variables considered as contributing factors to women’s likelihood to receive a vaccination (as reported in prior literature) were assessed and included as covariates in our analyses. These variables included a single item question assessing whether or not the participant was sexually active, whether or not they had health insurance, and whether or not they knew a friend who had gotten an HPV infection (these single item variables were dummy-coded; 0 = No, 1 = Yes). We also used single-item questions to assess perceived risk of acquiring HPV, asking participants to report “To what extent do you feel you are at risk for contracting HPV?” on a scale from 1 (*Not at All At Risk*) to 7 (*High Risk*), and perceptions of both the HPV vaccine’s safety and its effectiveness using a slider scale from 0 (*Not Safe/Effective At All*) to 100 (*Completely Safe/Effective*). Finally, we created a brief 10-item questionnaire to assess participants’ basic general knowledge about HPV. This questionnaire used a True/False format and included items such as “HPV can cause cervical cancer” and “men cannot get HPV.” Correct answers were summed to create a knowledge score out of 10 possible points.

Proposed analyses

In order to test Hypothesis 1, we will first conduct bivariate correlation analyses to establish if feminine honor endorsement is associated with decreased HPV vaccination intentions for currently unvaccinated college women, as manifested by a significant negative correlation between the HIW and our HPV intentions measurement. Next, in order to test Hypothesis 2 investigating the mediating roles played by HPV vaccination stigma and shame on the feminine honor-HPV intentions link, we will conduct sequential mediation analyses using Hayes’ PROCESS macro⁶¹ in SPSS. This will allow us to test our hypothesis using bootstrapped confidence intervals, a method which has proven to be quite robust in comparison to more traditional approaches to mediation analyses.⁶²

Results

Descriptive statistics and hypothesis 1

Bivariate correlations, means, and standard deviation can be found in Table 1. The HIW was correlated with higher levels of HPV-Stigma and HPV-Shame ($r_s > .18$, $p_s < .05$). Both HPV-Stigma and HPV-Shame were significantly

Table 1. Bivariate correlations, means, and standard deviations for the variables of interest.

Variable	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	M	SD
1. HIW	–	.30**	.18*	–.15*	–.10	.02	–.28**	–.22*	–.16*	.09	–.07	4.83	1.23
2. HPV-Stigma		–	.72**	–.29**	–.11	.10	–.36**	–.38**	–.29**	.11	–.13	2.34	1.32
3. HPV-Shame			–	–.34**	–.15*	–.01	–.32**	–.35**	–.21**	.04	–.09	2.64	1.53
4. HPV-Intent				–	.19*	.25**	.43**	.37**	.18*	.01	.01	4.95	1.70
5. Active					–	.34**	–.10	–.05	.10	–.11	–.06	.46	.50
6. Risk						–	–.05	–.05	.20*	–.02	.04	2.47	1.44
7. Safety							–	.79**	.20**	–.01	.06	76.03	21.10
8. Effect								–	.22**	–.03	.02	74.38	19.58
9. Knowledge									–	–.08	.09	6.55	1.25
10. Insure										–	–.03	.92	.27
11. Friend											–	.06	.23

Note: Ns range from 176 to 181 due to missingness. ** $p < .01$; * $p < .05$. HIW = Honor Ideology for Womanhood Scale. Active = Sexually Active. Risk = Perceived Risk. Safety = Perception of HPV Vaccine Safety. Effect = Perception of HPV Vaccine Efficacy. Knowledge = HPV Knowledge Score. Insure = Insurance Status. Friend = Knowing a Friend Who Had Contracted an HPV Infection.

correlated with HPV vaccination intentions ($r_s = -.29$ and $-.34$, respectively, $p_s < .01$). HPV knowledge was moderately high ($M_{\text{score}} = 6.55$, $SD = 1.25$), and knowledge was correlated with higher HPV vaccination intentions ($r = .18$, $p < .05$). Perception of safety, effectiveness, and risk were all correlated with higher HPV vaccination intentions ($r_s > .25$, $p_s < .01$). Finally, providing support for Hypothesis 1, analyses revealed that the HIW was correlated with lower levels of HPV vaccination intentions ($r = -.15$, $p = .04$).

Hypothesis 2

In order to test if feminine honor impacted HPV vaccination intentions indirectly through HPV-Stigma and HPV-Shame and embarrassment, we ran sequential mediation analyses using the PROCESS macro⁶¹ in SPSS (Model 6) via path analysis and using the percentile bootstrapping method of 95% confidence intervals. More specifically, we tested the indirect effect using 5,000 bootstrapped samples to produce confidence intervals testing the indirect effect—intervals are significant at $p < .05$ if the interval does not contain a coefficient of 0. Sequential mediation analyses testing the impact of feminine honor endorsement on HPV vaccination intentions through sexual purity stigma and HPV vaccination shame and embarrassment were conducted while controlling for whether or not the participant was sexually active, had health insurance, and knew a friend who had gotten an HPV infection. In these analyses, we also controlled for perceived risk, perceptions of the HPV vaccine's safety and effectiveness, and HPV knowledge. The analyses revealed a significant 3-path mediated effect of the HIW on HPV-Intent through HPV-Stigma and HPV-Shame (Mediated Effect-ME = $-.03$, $SE = .02$, 95% CI = $[-.093, -.002]$, $p < .05$). This path demonstrates that higher feminine honor endorsement is linked with higher levels of HPV vaccination sexual purity stigma, which is linked with higher HPV vaccination shame and embarrassment, which is subsequently linked with decreased HPV vaccination intentions. The two-path mediated effect of the HIW on HPV-Intent through HPV-Stigma was not significant (ME = $.004$, $SE = .03$, 95% CI = $[-.049, .065]$, $p > .05$). The two-path mediated

effect of the HIW on HPV-Intent through HPV-Shame was not significant either (ME = $.02$, $SE = .02$, 95% CI = $[-.017, .03]$, $p > .05$). The final model can be found in Figure 1. Consistent with our hypothesis, these analyses revealed that feminine honor endorsement significantly impacts HPV vaccination intentions through the sexual promiscuity stigma of HPV vaccination and the anticipated shame and embarrassment a woman would feel if it was known that she had received an HPV vaccination.

General discussion

The focus of the CDC, in regard to HPV vaccination, has aimed at two populations. The first group corresponds to children between 9 and 12 years old who have been targeted in order to protect them against HPV infections and HPV-related diseases prior to the beginning of sexual contact. The second targeted group has been young adult women aged 26 years old or younger who have not been vaccinated in the past or who have not completed all three doses of the HPV vaccine in their younger age. This second group of women should continue to be an essential target for HPV vaccination, given their high prevalence of HPV infection. More important, when taking into account the relatively low uptake of HPV vaccination in the U.S., it is necessary to determine which factors might prevent individuals in this “catch-up” vaccination group from receiving the HPV vaccination. The current study intended to address how feminine honor endorsement, which emphasizes the importance of maintaining a reputation of sexual purity and fidelity, may decrease young women's intention to receive the HPV vaccine. Using a sample of young college women, we found support for our first hypothesis. The results of the study showed that feminine honor endorsement was linked with lower levels of HPV vaccination intentions. The analyses also revealed a significant indirect effect of feminine honor endorsement on HPV vaccination intentions through HPV vaccination sexual purity stigma and the anticipated shame and embarrassment women would feel if others knew they had received the vaccine, providing support for our second hypothesis.

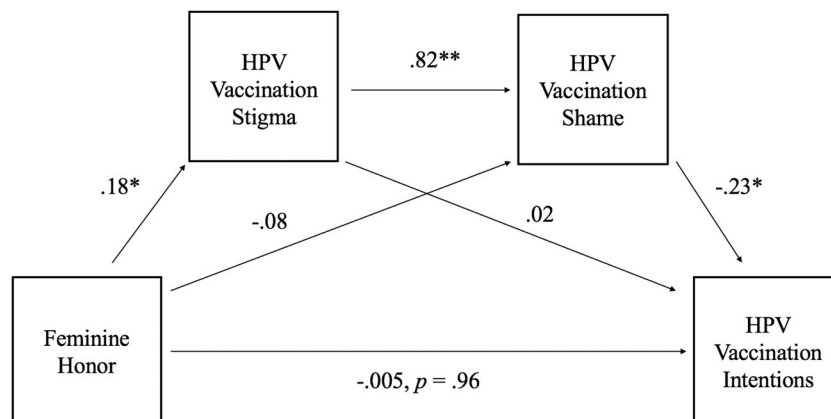


Figure 1. The impact of feminine honor on HPV vaccination intentions through HPV vaccination stigma and HPV vaccination shame. Note: $N = 157$. $^{**} p < .01$; $^* p < .05$. All coefficients unstandardized.

These findings contribute to an emerging body of literature which supports the view that feminine honor endorsement decreases the likelihood of women to seek out certain procedures and vaccinations which may have attached to them the stigma of sexual promiscuity.^{48,63} The findings of our study also contribute to research showing how pernicious attitudes prevent HPV vaccination by demonstrating how a widespread cultural factor can amplify beliefs that receiving an HPV vaccination implies some form of sexual promiscuity.^{30–36} It should be noted that our findings were found in a sample of college women, a cohort that has some of the highest rates of HPV infection.⁴ Accordingly, finding ways in which to minimize the sexual purity stigma attached to HPV vaccinations may help alleviate the impact of feminine honor endorsement on HPV vaccination intentions. One way to minimize such stigma is to educate college women that HPV infection has a well-established connection with various preventable cancers, and HPV vaccination is the primary preventive measure to reduce such incidence. It may also be beneficial to encourage college women to view HPV vaccination as a common preventive and safety measure (e.g., seatbelts or helmets wearing), or as one of many ordinary immunizations needed for protection from onset of certain diseases (e.g., diphtheria). These approaches may help reduce the stigma rooted from concerns about sexual promiscuity. Presenting the HPV vaccination as a means of lifesaving and dismissing its connection to sexual activity would be also helpful to minimize the sexual purity stigma in these young adult women.

It should also be noted that one would not expect sexual purity stigma to influence honor-endorsing men, as masculine reputation in honor cultures is primarily centered around strength and virility.^{52,64,65} However, it is possible that a similar stigma may be influencing honor-endorsing men as well. For example, there is preliminary evidence supporting the view that men may hesitate to seek out the HPV vaccine because of HPV infection initially being portrayed as a “feminine” infection. Due to this, some men have indicated that receiving the HPV vaccine would undermine their masculinity.⁶⁶ If this is a prevalent perspective, one might suspect that honor-endorsing men would have particularly strong aversion to receiving an HPV vaccination. Future research should investigate the possibility of masculine honor decreasing HPV vaccination intentions in men as well.

One might consider the possibility that, for honor-endorsing women, levels of stigma may be lower in women who have previously received the vaccine. After all, prior research shows that receiving/partaking in a procedure/screening tends to minimize negative attitudes toward the procedure/screening.^{67–69} However, data that we have collected in a separate sample reveals that the link between feminine honor endorsement and HPV sexual purity stigma is not attenuated in women who reported previously having received the vaccine.⁷⁰ This inconsistency with prior research emphasizes the apparent strength of feminine honor endorsement, in that even individuals who had previously received the HPV vaccine still appear to endorse beliefs that receiving the vaccine implies that a woman partakes in sexually-promiscuous

behavior. This has a few possible implications. For example, women who have received the vaccine may not feel that the negative attitudes they endorse apply to themselves, only to other women choosing to receive the vaccine. This would be consistent with attribution theory^{71–73}, which posits that we are predisposed to attributing our own negative behaviors to external circumstances. In this case, perhaps women attribute their decision to receive the HPV vaccination to less controllable factors (i.e. higher risk due to family history of cervical cancer or the vaccination decision was imposed on them by their parents) as opposed to their own sexual behavior. Another explanation is that honor-endorsing women still hold these negative attitudes about themselves for receiving the HPV vaccine and are perhaps engaging in a form of self-stigmatization.⁷⁴ This would be problematic considering the mental health outcomes shown to be associated with self-stigmatization, such as psychological distress, anxiety, and depression.^{75–77} Future research should further explore the negative attitudes of honor-endorsing women who have received the HPV vaccine.

One limitation of the current study is that only behavioral intentions were measured, as opposed to actual commitment to receiving an HPV vaccine. Due to the multi-dose nature of an HPV vaccination, and the fact that each dose is separated by multiple months, it would take the setup of a longitudinal study in order to measure adherence to the traditional 3-wave vaccination in young women. However, one might consider the Health Belief Model⁷⁸ as support for analyzing behavioral intentions in this instance. Perceived barriers, otherwise defined as the beliefs regarding “the tangible and psychological costs of the advised action”, have been shown to be the strongest predictor of actual health behavior.⁷⁹ Given that the current study was centered around how feminine honor might heighten perceived stigma (e.g. beliefs that there would be social repercussions for receiving the vaccine), it is likely that the current study’s measurements were the most appropriate approach within a cross-sectional design.

Overall, there are other significant barriers to care in regard to HPV vaccination. However, to our knowledge, this study is the first to address how feminine honor, a common cultural ideology in the U.S., may heighten HPV vaccination stigma in college women, and prevent them from seeking out such a vaccination. Through continued research that focus on identifying factors which might help to attenuate these effects, it is possible that many more college women might be successfully targeted in the “catch-up” HPV vaccination phase.

Conflict of interest disclosure

The authors have no conflicts of interest to report. The authors confirm that the research presented in this article met the ethical guidelines, including adherence to the legal requirements, of the United States of America and received approval from the Institutional Review Board of University of Oklahoma.

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References

- Center for Disease Control. 2019a. Genital HPV Infection—Fact Sheet. Retrieved from <https://www.cdc.gov/std/hpv/stdfact-hpv.htm>.
- Smith JS, Melendy A, Rana RK, Pimenta JM. Age-specific prevalence of infection with human papillomavirus in females: a global review. *J Adolesc Health*. 2008;43(4):S5-e1–S5.e62. doi:10.1016/j.jadohealth.2008.07.009.
- Dunne EF, Unger ER, Sternberg M, et al. Prevalence of HPV infection among females in the United States. *J Am Med Assoc*. 2007;297(8):813–819. doi:10.1001/jama.297.8.813.
- Thompson EL, Vámos CA, Vázquez-Otero C, Logan R, Griner S, Daley EM. Trends and predictors of HPV vaccination among US College women and men. *Prev Med*. 2016;86:92–98. doi:10.1016/j.ypmed.2016.02.003.
- Center for Disease Control. 2019b. HPV-Associated cancer statistics. Retrieved from <https://www.cdc.gov/cancer/hpv/statistics/index.htm>.
- Center for Disease Control. 2019c. Vaccinating boys and girls. Retrieved from <https://www.cdc.gov/hpv/parents/vaccine.html>.
- Walker TY, Elam-Evans LD, Yankey D, et al. National, regional, state, and selected local area vaccination coverage among adolescents aged 13–17 years—United States, 2018. *MMWR Morb Mortal Wkly Rep*. 2019;68(33):718–723. doi:10.15585/mmwr.mm6833a2externalicon.
- Meites E, Szilagyi PG, Chesson HW, Unger ER, Romero JR, Markowitz LE. Human papillomavirus vaccination for adults: Updated recommendations of the Advisory Committee on Immunization Practices. *Am J Transplant*. 2019;19(11):3202–3206. doi:10.1111/ajt.15633.
- Westra TA, Rozenbaum MH, Rogoza RM, et al. Until which age should women be vaccinated against HPV infection? Recommendation based on cost-effectiveness analyses. *J Infect Dis*. 2011;204(3):377–384. doi:10.1093/infdis/jir281.
- Oshman LD, Davis AM. Human papillomavirus vaccination for adults: Updated recommendations of the Advisory Committee on Immunization Practices (ACIP). *JAMA*. 2020;323(5):468–469. doi:10.1001/jama.2019.18411.
- Castellsague X, Munoz N, Pitisuttithum P, et al. End-of-study safety, immunogenicity, and efficacy of quadrivalent HPV (types 6, 11, 16, 18) recombinant vaccine in adult women 24–45 years of age. *Br J Cancer*. 2011;105(1):28–37. doi:10.1038/bjc.2011.185.
- Luna J, Plata M, Gonzalez M, et al. Long-term follow-up observation of the safety, immunogenicity, and effectiveness of Gardasil™ in adult women. *PloS One*. 2013;8(12):e83431. doi:10.1371/journal.pone.0083431.
- Skinner SR, Szarewski A, Romanowski B, et al. Efficacy, safety, and immunogenicity of the human papillomavirus 16/18 AS04-adjuvanted vaccine in women older than 25 years: 4-year interim follow-up of the phase 3, double-blind, randomised controlled VIVIANE study. *The Lancet*. 2014;384(9961):2213–2227. doi:10.1016/S0140-6736(14)60920-X.
- Wheeler CM, Skinner SR, Del Rosario-Raymundo MR, et al. Efficacy, safety, and immunogenicity of the human papillomavirus 16/18 AS04-adjuvanted vaccine in women older than 25 years: 7-year follow-up of the phase 3, double-blind, randomised controlled VIVIANE study. *The Lancet Infect Dis*. 2016;16(10):1154–1168. doi:10.1016/S1473-3099(16)30120-7.
- Williams WW, Lu P-J, Saraiya M, et al. Factors associated with human papillomavirus vaccination among young adult women in the United States. *Vaccine*. 2013;31(28):2937–2946. doi:10.1016/j.vaccine.2013.04.041.
- Boersma P, Black LI. Human papillomavirus vaccination among adults aged 18–26, 2013–2018. 2020.
- Manhart LE, Burgess-Hull AJ, Fleming CB, Bailey JA, Haggerty KP, Catalano RF. HPV vaccination among a community sample of young adult women. *Vaccine*. 2011;29(32):5238–5244. doi:10.1016/j.vaccine.2011.05.024.
- Marchand E, Glenn BA, Bastani R. Low HPV vaccine coverage among female community college students. *J Commun Health*. 2012;37(6):1136–1144. doi:10.1007/s10900-012-9572-x.
- Rambout L, Tashkandi M, Hopkins L, Tricco AC. Self-reported barriers and facilitators to preventive human papillomavirus vaccination among adolescent girls and young women: a systematic review. *Prev Med*. 2014;58:22–32. doi:10.1016/j.ypmed.2013.10.009.
- Bednarczyk RA, Birkhead GS, Morse DL, Doleyres H, McNutt LA. Human papillomavirus vaccine uptake and barriers: association with perceived risk, actual risk and race/ethnicity among female students at a New York State university, 2010. *Vaccine*. 2011;29(17):3138–3143. doi:10.1016/j.vaccine.2011.02.045.
- Liddon NC, Hood JE, Leichter JS. Intent to receive HPV vaccine and reasons for not vaccinating among unvaccinated adolescent and young women: findings from the 2006–2008 National Survey of Family Growth. *Vaccine*. 2012;30(16):2676–2682. doi:10.1016/j.vaccine.2012.02.007.
- Thompson EL, Vámos CA, Piepenbrink R, et al. Human papillomavirus risk perceptions and relationship status: a barrier to HPV vaccination? *J Behav Med*. 2019;42(5):991–997. doi:10.1007/s10865-019-00025-4.
- Patel DA, Zochowski M, Peterman S, Dempsey AF, Ernst S, Dalton VK. Human papillomavirus vaccine intent and uptake among female college students. *J Am Coll Health*. 2012;60(2):151–161. doi:10.1080/07448481.2011.580028.
- Bendik MK, Mayo RM, Parker VG. Knowledge, perceptions, and motivations related to HPV vaccination among college women. *J Canc Educ*. 2011;26(3):459–464. doi:10.1007/s13187-011-0200-8.
- Chao C, Velicer C, Slezak JM, Jacobsen SJ. Correlates for human papillomavirus vaccination of adolescent girls and young women in a managed care organization. *Am J Epidemiol*. 2010;171(3):357–367. doi:10.1093/aje/kwp365.
- Read DS, Joseph MA, Polishchuk V, Suss AL. Attitudes and perceptions of the HPV vaccine in Caribbean and African-American adolescent girls and their parents. *J Pediatr Adolesc Gynecol*. 2010;23(4):242–245. doi:10.1016/j.jpaa.2010.02.002.
- Zimet GD, Weiss TW, Rosenthal SL, Good MB, Vichnin MD. Reasons for non-vaccination against HPV and future vaccination intentions among 19–26 year-old women. *BMC Women's Health*. 2010;10(1):27. doi:10.1186/1472-6874-10-27.
- Thompson EL, Wheldon CW, Rosen BL, Maness SB, Kasting ML, Massey PM. Awareness and knowledge of HPV and HPV vaccination among adults ages 27–45 years. *Vaccine*. 2020;38(15):3143–3148. doi:10.1016/j.vaccine.2020.01.053.
- Licht AS, Murphy JM, Hyland AJ, Fix BV, Hawk LW, Mahoney MC. Is use of the human papillomavirus vaccine among female college students related to human papillomavirus knowledge and risk perception? *Sexually Transmitted Infect*. 2010;86(1):74–78. doi:10.1136/sti.2009.037705.
- Hopfer S, Clippard JR. College women's HPV vaccine decision narratives. *Qual Health Res*. 2011;21(2):262–277. doi:10.1177/1049732310383868.
- Wang S-M, Zhang S-K, Pan X-F, et al. Human papillomavirus vaccine awareness, acceptability, and decision-making factors among Chinese college students. *Asian Pacific J Cancer Prev*. 2014;15(7):3239–3245. doi:10.7314/APJCP.2014.15.7.3239.
- Pierre-Victor D, Stephens DP, Omondi A, Clarke R, Jean-Baptiste N, Madhivanan P. Barriers to HPV Vaccination among unvaccinated, Haitian American college women. *Health Equity*. 2018;2(1):90–97. doi:10.1089/heq.2017.0028.

33. Loke AY, Chan ACO, Wong YT. Facilitators and barriers to the acceptance of human papillomavirus (HPV) vaccination among adolescent girls: a comparison between mothers and their adolescent daughters in Hong Kong. *BMC Res Notes*. 2017;10(1):390. doi:10.1186/s13104-017-2734-2.
34. Williams K, Forster A, Marlow L, Waller J. Attitudes towards human papillomavirus vaccination: a qualitative study of vaccinated and unvaccinated girls aged 17–18 years. *J Family Plan Reprod Health Care*. 2011;37(1):22–25. doi:10.1136/jfprhc.2010.0017.
35. Wong LP. Young multiethnic women's attitudes toward the HPV vaccine and HPV vaccination. *Int J Gynecol Obstetrics*. 2008;103(2):131–135. doi:10.1016/j.ijgo.2008.07.005.
36. Krakow MM, Jensen JD, Carcioppolo N, Weaver J, Liu M, Guntzville LM. Psychosocial predictors of human papillomavirus vaccination intentions for young women 18 to 26: religiosity, morality, promiscuity, and cancer worry. *Women's Health Issues*. 2015;25(2):105–111. doi:10.1016/j.whi.2014.11.006.
37. Brown RP. 2016. *Honor Bound: How a Cultural Ideal Has Shaped the American Psyche*. New York, NY: Oxford University Press.
38. Cohen D, Nisbett RE. Self-protection and the culture of honor: Explaining southern violence. *Pers Soc Psychol Bull*. 1994;20(5):551–567. doi:10.1177/0146167294205012.
39. Cross SE, Uskul AK, Gerçek-Swing B, Alözkan C, Ataca B. Confrontation versus withdrawal: Cultural differences in responses to threats to honor. *Group Processes Intergroup Relat*. 2013;16(3):345–362. doi:10.1177/1368430212461962.
40. Mosquera PMR, Fischer AH, Manstead AS. *Inside the Heart of Emotion: On Culture and Relational Concerns*. LZ Tiedens and CW Leach (eds). Cambridge: Cambridge University Press; 2004: 187–202.
41. Saucier DA, McManus JL. Men of honor: Examining individual differences in masculine honor beliefs. In *Masculinities in a Global Era*. New York, NY: Springer; 2014:85–99.
42. Vandello JA, Cohen D, Grandon R, Franiuk R. Stand by your man: Indirect prescriptions for honorable violence and feminine loyalty in Canada, Chile, and the United States. *J Cross-Cult Psychol*. 2009;40(1):81–177. doi:10.1002/2022107313862.
43. Wyatt-Brown B. *Southern Honor: Ethics and Behavior in the Old South*. Oxford: Oxford University Press; 1982.
44. Cohen D, Nisbett RE, Bowdle BF, Schwarz N. Insult, aggression, and the southern culture of honor: an experimental ethnography. *J Personality Soc Psychol*. 1996;70(5):945–959. doi:10.1037/0022-3514.70.5.945.
45. Brown RP, Osterman LL, Barnes CD. School violence and the culture of honor. *Psychol Sci*. 2009;20(11):1400–1405. doi:10.1111/j.1467-9280.2009.02456.x.
46. Grosjean P. A history of violence: The culture of honor and homicide in the US South. *J Euro Econ Assoc*. 2014;12(5):1285–1316. doi:10.1111/jeea.12096.
47. Saucier DA, Till DE, Miller SS, O'Dea CJ, Andres E. Slurs against masculinity: Masculine honor beliefs and men's reactions to slurs. *Language Sci*. 2015;52:108–120. doi:10.1016/j.langsci.2014.09.006.
48. Foster SD, Carvalho MR, Song H, Lee J, Fisher R, Traxler H. An implication of impurity: The impact of feminine honor on HPV screenings and the decision to authorize daughter's HPV vaccinations. *Stigma Health*. 2021;6(2):216–227. doi:10.1037/SAH-2019-0121R1.
49. Rodriguez Mosquera PM. On the importance of family, morality, masculine, and feminine honor for theory and research. *Soc Personal Psychol Compass*. 2016;10(8):431–442. doi:10.1111/spc3.12262.
50. Brown RP, Imura M, Mayeux L. Honor and the stigma of mental healthcare. *Personal Soc Psychol Bull*. 2014;40(9):1119–1131. doi:10.1177/0146167214536741.
51. Guerra VM, Giner-Sorolla R, Vasiljevic M. The importance of honor concerns across eight countries. *Group Processes Intergroup Relat*. 2013;16(3):298–318. doi:10.1177/1368430212463451.
52. Mosquera PMR. Masculine and feminine honor codes. *Revista de Psicol Soc*. 2011;26(1):63–72. doi:10.1174/021347411794078499.
53. Vandello JA, Cohen D, Ransom S. US Southern and Northern differences in perceptions of norms about aggression: Mechanisms for the perpetuation of a culture of honor. *J Cross-Cult Psychol*. 2008;39(2):162–177. doi:10.1177/0022022107313862.
54. Brown RP, Baughman K, Carvalho M. Culture, masculine honor, and violence toward women. *Personality Soc Psychol Bull*. 2018;44(4):538–549. doi:10.1177/0146167217744195.
55. Dietrich DM, Schuett JM. Culture of honor and attitudes toward intimate partner violence in Latinos. *Sage Open*. 2013;3(2): 215824401348968. doi:10.1177/2158244013489685.
56. Vandello JA, Cohen D. Male honor and female fidelity: implicit cultural scripts that perpetuate domestic violence. *J Personal Soc Psychol*. 2003;84(5):997. doi:10.1037/0022-3514.84.5.997.
57. Gul P, Cross SE, Uskul AK. Implications of culture of honor theory and research for practitioners and prevention researchers. *Am Psychologist*. Advance Online Publication. 2020; doi:10.1037/amp0000653.
58. Cohen D. Culture, social organization, and patterns of violence. *J Personal Soc Psychol*. 1998;75(2):408–419. doi:10.1037/0022-3514.75.2.408.
59. Fortenberry JD, McFarlane M, Bleakley A, et al. Relationships of stigma and shame to gonorrhea and HIV screening. *Am J Public Health*. 2002;92(3):378–381. doi:10.2105/AJPH.92.3.378.
60. Barnes CD, Brown RP, Lenes J, Bosson J, Carvalho M. My country, my self: Honor, identity, and defensive responses to national threats. *Self Identity*. 2014;13(6):638–662. doi:10.1080/15298868.2014.892529.
61. Hayes AE. *Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach*. New York, NY: Guilford Publications; 2017.
62. Preacher KJ, Hayes AE. Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behav Res Methods*. 2008;40(3):879–891. doi:10.3758/BRM.40.3.879.
63. Foster SD, Carvalho MR, Lee J. Feminine honor endorsement and women's STI screenings: An observational study. *Psychol Health Med*. 2021;1–6. doi:10.1080/13548506.2021.1883693.
64. Canto JM, Alvaro JL, Pereira C, Garrido A, Torres AR, Pereira ME. Jealousy, gender, and culture of honor: A study in Portugal and Brazil. *J Psychol*. 2017;151(6):580–596. doi:10.1080/00223980.2017.1372344.
65. López-Zafra E. Relación entre Cultura del Honor e Identidad de género: el papel del sexo, edad y nivel de estudios en la predisposición a la violencia. *Estudios de Psicol*. 2008;29(2):209–220. doi:10.1174/021093908784485101.
66. Nadarzynski T, Smith H, Richardson D, Pollard A, Llewellyn C. Perceptions of HPV and attitudes towards HPV vaccination amongst men who have sex with men: A qualitative analysis. *Br J Health Psychol*. 2017;22(2):345–361. doi:10.1111/bjhp.12233.
67. Heimberger T, Chang HG, Shaikh M, Crotty L, Morse D, Birkhead G. Knowledge and attitudes of healthcare workers about influenza: why are they not getting vaccinated? *Infect Control Hospital Epidemiol*. 1995;16(7):412–415. <https://doi-org.ezproxy.lib.ou.edu/10.2307/30141897> doi:10.2307/30141897.
68. Nguyen TT, McPhee SJ, Nguyen T, Lam T, Mock J. Predictors of cervical Pap smear screening awareness, intention, and receipt among Vietnamese-American women. *Am J Prev Med*. 2002;23(3):207–214. doi:10.1016/S0749-3797(02)00499-3.
69. Sung JF, Blumenthal DS, Coates RJ, Alema-Mensah E. Knowledge, beliefs, attitudes, and cancer screening among inner-city African-American women. *J Natl Med Assoc*. 1997;89(6):405–411.
70. Foster SD, Carvalho MR. 2020. [Feminine honor by vaccination status interaction]. Unpublished raw data.
71. Böhm G, Pfister HR. How people explain their own and others' behavior: a theory of lay causal explanations. *Front Psychol*. 2015;6:139. doi:10.3389/fpsyg.2015.00139.
72. Heider F. *The Psychology of Interpersonal Relations*. New York, NY: Psychology Press; 1958.
73. Weiner B. An attributional theory of achievement motivation and emotion. *Psychol Rev*. 1985;92(4):548–573. doi:10.1037/0033-295X.92.4.548.

74. Bathje GJ, Marston HN. Self-stigmatization. *Encyclopedia of Critical Psychology*. New York: Springer; 2014: 1713–1716. doi:[10.1007/978-1-4614-5583-7](https://doi.org/10.1007/978-1-4614-5583-7).
75. Herek GM, Saha S, Burack J. Stigma and psychological distress in people with HIV/AIDS. *Basic Appl Soc Psychol*. 2013;35(1):41–54. doi:[10.1080/01973533.2012.746606](https://doi.org/10.1080/01973533.2012.746606).
76. Lysaker PH, Yanos PT, Outcalt J, Roe D. Association of stigma, self-esteem, and symptoms with concurrent and prospective assessment of social anxiety in schizophrenia. *Clin Schizophrenia Relat Psychoses*. 2010;4(1):41–48. doi:[10.3371/CSRP.4.1.3](https://doi.org/10.3371/CSRP.4.1.3).
77. Schomerus G, Corrigan PW, Klauer T, Kuwert P, Freyberger HJ, Lucht M. Self-stigma in alcohol dependence: consequences for drinking-refusal self-efficacy. *Drug Alcohol Depend*. 2011;114(1):12–17. doi:[10.1016/j.drugalcdep.2010.08.013](https://doi.org/10.1016/j.drugalcdep.2010.08.013).
78. Rosenstock IM. Historical origins of the health belief model. *Health Educ Monographs*. 1974;2(4):328–335. doi:[10.1177/109019817400200403](https://doi.org/10.1177/109019817400200403).
79. Champion VL, Skinner CS. The health belief model. *Health Behav Health Educ Theory Res Pract*. 2008;4:45–65.